

Parsing Programming Languages

Sujit Kumar Chakrabarti

IIITB

Parsing

- ▶ Process of determining if the input belongs to the language of the grammar
- ▶ On success – builds a *parse tree*

Parsing

Grammar

Example: Grammar for arithmetic expressions

Parsing

Grammar

Language of grammar

$L(G)$

Backus Naur Form (BNF).

::=

Example: Grammar for arithmetic expressions
Grammar strings.

E
 E
 E
 E

$\rightarrow$
 $\rightarrow$
 $\rightarrow$
 $\rightarrow$

$E + E$
 $E * E$
 (E)
 $0 \mid 1 \dots \mid 9$

Rule, production.

Terminals / Tokens.

GrammarSymbol.

Non
terminals -

Parsing

Grammar – Activity

Example: Grammar for function call

Parsing

Grammar – Activity

Example: Grammar for function call

<i>stmt</i>	→	... <i>func-call</i> ; ...
<i>func – call</i>	→	id (arglist)
<i>arglist</i>	→	<i>arg</i> *
<i>arg</i>	→	<i>exp</i>
<i>exp</i>	→	... SL ...

Parsing

Derivation

Parsing

Derivation

Example: Derivations for arithmetic expressions

▶ 5

▶ $1 + 2$

▶ $1 + 2 * 3$

Parsing

Parse Tree

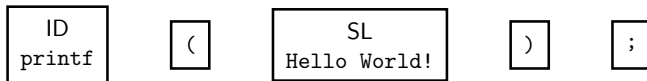
```
printf("Hello World!");
```

Parsing

Parse Tree

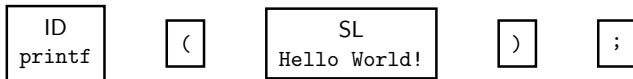
```
printf("Hello World!");
```

After lexical analysis:



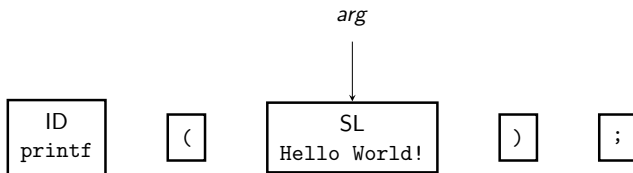
Parsing

Parse Tree



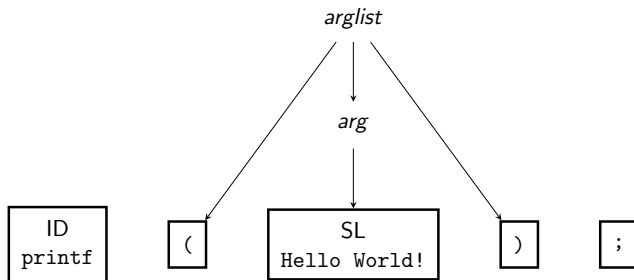
Parsing

Parse Tree



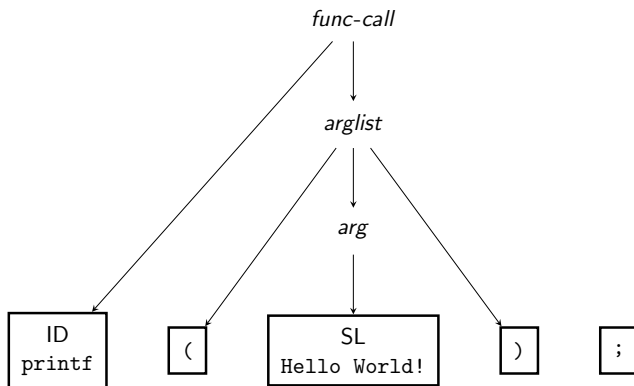
Parsing

Parse Tree



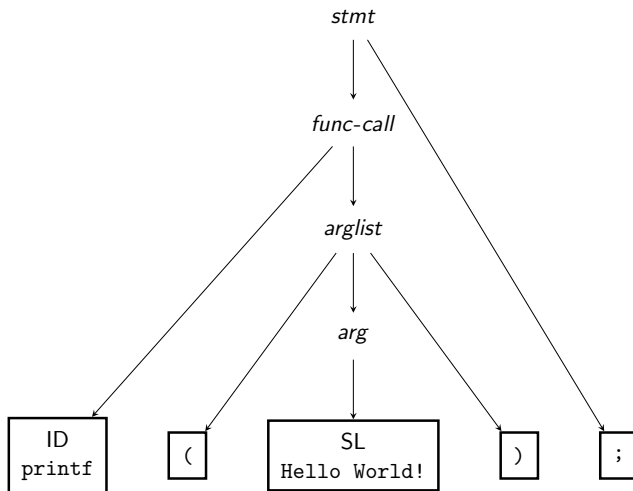
Parsing

Parse Tree



Parsing

Parse Tree



Parsing

Parse Tree $\rightarrow$ Grammar.

- ▶ Grammar symbol $\mapsto$ Nodes
- ▶ Starting symbol $\mapsto$ Root node
- ▶ Non-terminals $\mapsto$ internal nodes
- ▶ Terminals $\mapsto$ leaves
- ▶ Productions $\mapsto$ Edges

Parsing

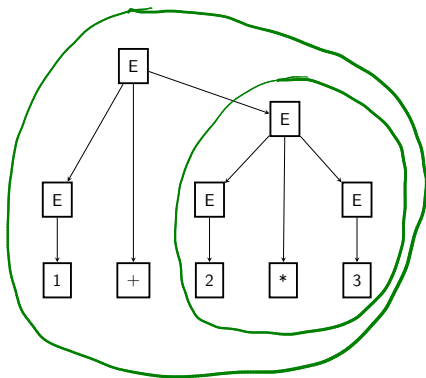
Ambiguity

$$1 + 2 * 3$$

Parsing

Ambiguity

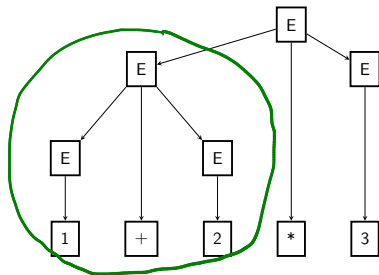
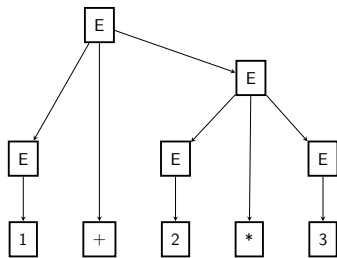
1 + 2 * 3



Parsing

Ambiguity

1 + 2 * 3



Parsing

Ambiguity – Methods of dealing with it

No parser for an ambiguous grammar

- ▶ Associativity
- ▶ Operator precedence

Parsing

Grammar – Activity

Example: Grammar for function call

Parsing

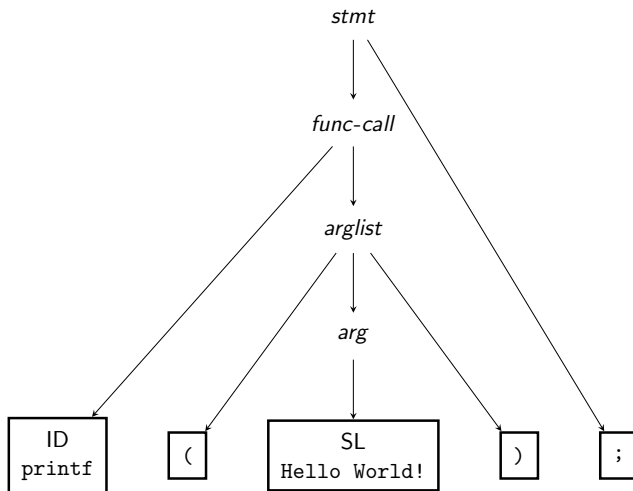
Grammar – Activity

Example: Grammar for function call

<i>stmt</i>	→	... <i>func-call</i> ; ...
<i>func – call</i>	→	id (arglist)
<i>arglist</i>	→	<i>arg</i> *
<i>arg</i>	→	<i>exp</i>
<i>exp</i>	→	... SL ...

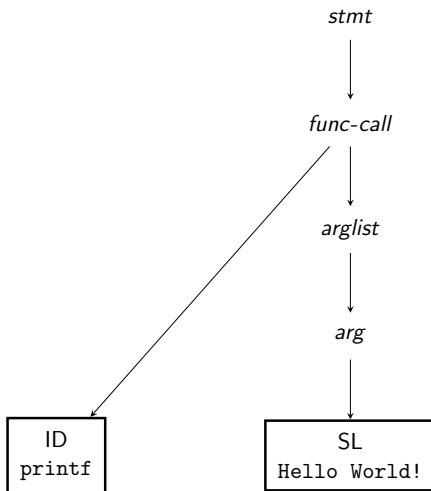
Parsing

Parse Tree



Parsing

Parse Tree



Parsing

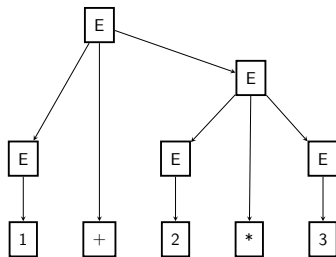
Abstract Syntax Tree

$1 + 2 * 3$

Parsing

Abstract Syntax Tree

$1 + 2 * 3$

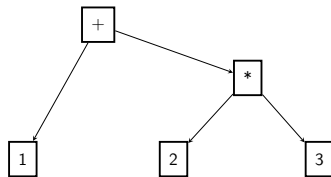
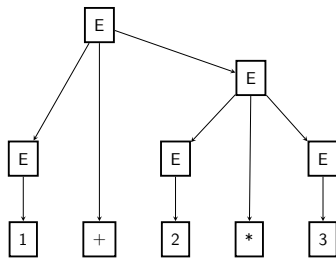


Parsing

Abstract Syntax Tree

$1 + 2 * 3$

Syntax
— Concrete
— Abstract



Recursive Descent Parsing

Recursive Descent Parsing

Derivation: Deriving ~~the~~ input string from the start symbol of the grammar.

- Leftmost
- rightmost

- ▶ Top-down parsing
- ▶ Starts with the root
- ▶ Prepares the parse tree in pre-order depth first sequence
- ▶ Finds the leftmost derivation for a string

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

Rightmost Derivation

E

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$\underline{1 + (2 + 3)} \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$E \Rightarrow \textcircled{E} + E$$

Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \end{aligned}$$

Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \end{aligned}$$

Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \end{aligned}$$

Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + \textcolor{green}{(E)} + E \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \end{aligned}$$

Rightmost Derivation

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$\underline{1 + (2 + 3)} \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow \underline{E} + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$E$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$E \Rightarrow E + E$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow E + (E) \end{aligned}$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow E + (E) \\ &\Rightarrow E + (E + E) \end{aligned}$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow E + (E) \\ &\Rightarrow E + (E + E) \\ &\Rightarrow E + (E + \text{id}_3) \end{aligned}$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow E + (E) \\ &\Rightarrow E + (E + E) \\ &\Rightarrow E + (E + \text{id}_3) \\ &\Rightarrow E + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

Grammar:

$$E \rightarrow E + E \mid (E) \mid \text{id}$$

Example:

$$1 + (2 + 3) \rightarrow \text{id}_1 + \text{id}_2 + \text{id}_3$$

Leftmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow \text{id}_1 + E \\ &\Rightarrow \text{id}_1 + (E) \\ &\Rightarrow \text{id}_1 + (E + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + E) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Rightmost Derivation

$$\begin{aligned} E &\Rightarrow E + E \\ &\Rightarrow E + (E) \\ &\Rightarrow E + (E + E) \\ &\Rightarrow E + (E + \text{id}_3) \\ &\Rightarrow E + (\text{id}_2 + \text{id}_3) \\ &\Rightarrow \text{id}_1 + (\text{id}_2 + \text{id}_3) \end{aligned}$$

Recursive Descent Parsing

Leftmost and Rightmost Derivation

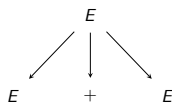
Formation of Parse Tree with Leftmost Derivation

E

Recursive Descent Parsing

Leftmost and Rightmost Derivation

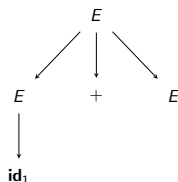
Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

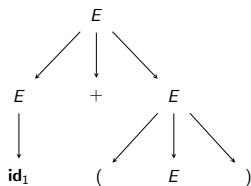
Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

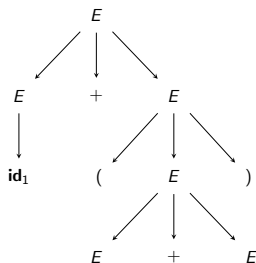
Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

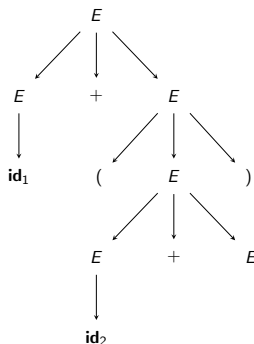
Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

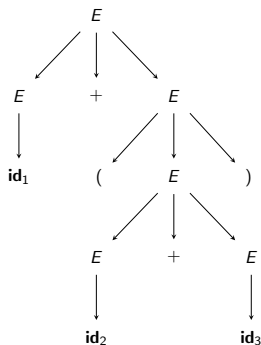
Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

Formation of Parse Tree with Leftmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

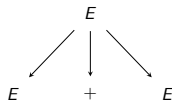
Formation of Parse Tree with Rightmost Derivation

E

Recursive Descent Parsing

Leftmost and Rightmost Derivation

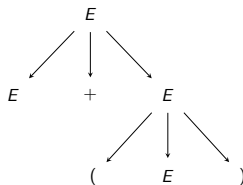
Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

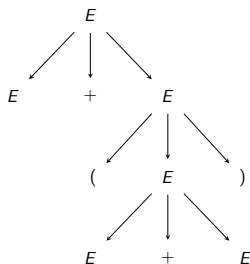
Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

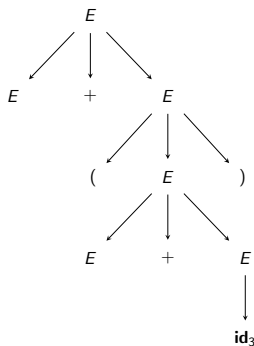
Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

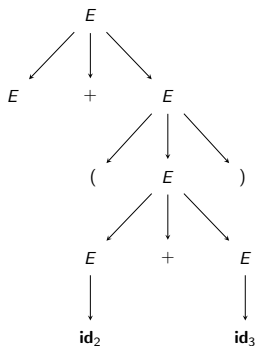
Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

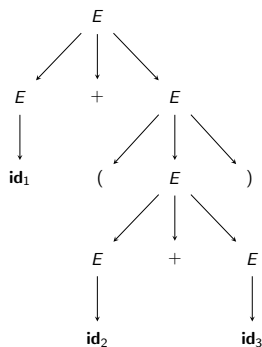
Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Leftmost and Rightmost Derivation

Formation of Parse Tree with Rightmost Derivation



Recursive Descent Parsing

Example

Grammar:

$$E \rightarrow \text{num } E'$$

$$E' \rightarrow + E E' \mid \epsilon$$

Example:

$$1 + 2 + 3 \rightarrow \text{num}_1 + \text{num}_2 + \text{num}_3$$

Recursive Descent Parsing

Algorithm

Grammar:

$$E \rightarrow \text{num } E'$$

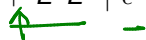
$$E' \rightarrow + E E' \mid \epsilon$$

Recursive Descent Parsing

Algorithm

Grammar:

$E \rightarrow \underline{\text{num}} E'$
 $E' \rightarrow + E E' \mid \epsilon$

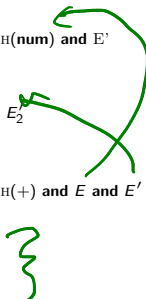


$\text{match}(t: \text{token})$

consume the next token t'
from the input

$t = t'$

```
procedure  $\dot{E}$   
  return MATCH(num) and  $E'$   
end procedure  
  
procedure  $E' \cdot$   
  return  $E'_1$  or  $E'_2$   
end procedure  
  
procedure  $E'_1$   
  return MATCH(+) and  $E$  and  $E'$   
end procedure  
  
procedure  $E'_2$   
  return true  
end procedure
```



Recursive Descent Parsing

Algorithm – Activity

Grammar:

$expr \rightarrow \mathbf{num} \mid term$
 $term \rightarrow factor \mid factor + term$
 $factor \rightarrow \mathbf{num} \mid \mathbf{num} * factor$

Recursive Descent Parsing

Left Recursion

Grammar:

$$\underline{E} \rightarrow \underline{E} + T \mid \text{num}$$

Recursive Descent Parsing

Left Recursion

$$E \rightarrow \underline{E} + E \mid \underline{E} * E$$

Leftmost derivation.
Left to right

Grammar:

$$E \rightarrow E + T \mid \text{num}$$

```
procedure E
  return E and MATCH(+) and MATCH(num)
end procedure
```

```
procedure T
  return E'_1 or E'_2
end procedure
```

Recursive Descent Parsing

Left Recursion

Grammar:

$$E \rightarrow E + T \mid \text{num}$$

```
procedure E
  return E and MATCH(+) and MATCH(num)
end procedure
```

```
procedure T
  return E1' or E2'
end procedure
```


yacc .

— Yet another compiler compiler.

ocaml yacc

LR ← rightmost
derivation
left to right

Donald Knuth .

→ Turing Award.

LaTeX